

Hunting Mr. X: Wellington Edition

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Abstract—

- **Problem:** hidden-movement board games need everyone in the same room, and existing digital versions use fictional boards.
- **Solution:** a browser-based, real-time multiplayer game played on a map of Wellington's real transport network.
- **Evaluation:** automated testing, a map-library benchmark, and two rounds of user testing with a usability questionnaire.
- **Outcome:** a deployed, playable system suitable for the university Open Day.

*Index Terms—*Multiplayer game, hidden movement, real-time web application, WebSocket, map overlay, usability evaluation.

I. INTRODUCTION

- Motivation: why hidden-movement games suit online play, and the gap this project fills (no browser version on a real city map).
- Secondary goal: a showpiece for the university Open Day.
- Aims and measurable goals, which the Evaluation section refers back to.
- Contributions, and a note on naming: the mechanics are used with the publisher's permission, and the original brand name is not used.
- Key findings and a roadmap of the report.

II. RELATED WORK

- The original board game and existing digital versions of it.
- Location-based games that use real-world maps.
- Keeping hidden information secret in online multiplayer games.
- Map libraries, and live routing services versus pre-computed routes [1].
- Real-time communication for web applications.
- Measuring usability with the System Usability Scale (SUS).
- How this project addresses the gaps these leave.

III. REQUIREMENTS AND CONSTRAINTS

- Functional requirements: the game rules the system has to enforce.
- Non-functional requirements: responsiveness, several games at once, and browser support.
- Constraints: no paid per-request services, no database, a solo project on a small budget, and self-hosting.
- A table linking each requirement to where it is evaluated.

This project was supervised by Jens Dietrich (primary) and Stuart Marshall.

IV. DESIGN

A. Architecture

- A client-server design in which the server is the single source of truth.
- The alternatives considered and why they were not chosen.

B. Game State Machine and Turn Flow

- How a game moves from the lobby through play to its end.
- What happens when a player leaves or stops responding.

C. Domain Model

- How players, tickets and rules are modelled, with the rules kept apart from the web framework so they can be tested directly.

D. Hidden Information and Role-Filtered State

- How each player is sent only what they are allowed to see, so Mr X's position stays secret.

E. Map Data Design

- Why the Wellington map is a pre-computed graph rather than live routing.
- How the map library was chosen.

F. User Interface Design

- The main screens, and the changes made in response to user feedback.

G. Real-World Issues

- Ethics, sustainability, safety, intellectual property and privacy.

V. IMPLEMENTATION

- Technology choices and the reasons for them.
- How the server enforces the rules and runs several games at once.
- How the browser client shows the map, the game state and live updates.
- Building the Wellington map, the tooling made for it, and the problems met along the way.
- Deployment, and keeping the API documentation in step with the code.
- The development process, including how AI-assisted development was used and checked.
- Screenshots of the main screens.

VI. EVALUATION

A. *Map Library Benchmark*

- How three map libraries were compared, the results, and the limits of the benchmark.

B. *Functional Correctness*

- The layers of automated tests and what each one checks.
- Regression tests for the bugs found in the backend review.
- Coverage, mutation and property-based testing as checks on the tests themselves.

C. *Performance and Multiplayer Synchronisation*

- Update latency and map rendering, measured against the targets from the Introduction.

D. *User Evaluation*

- Two rounds of user testing and the SUS questionnaire.
- The main themes in the feedback and the changes they led to.
- Limitations of the study.

E. *Requirements Traceability*

- Each goal from the Introduction marked as met, partly met or not met.

VII. DISCUSSION

- What worked well and what was hard.
- Lessons learned, including a security weakness found in review and how it was fixed.
- Trade-offs revisited, and threats to validity.

VIII. CONCLUSIONS AND FUTURE WORK

- What was delivered and the main results.
- Future work for a following student: a tutorial, configurable lobbies, saved games and accounts, mobile support, automated map generation, balance testing with bot players, and a larger user study.

ACKNOWLEDGMENTS

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REFERENCES

- [1] A. Author, "Placeholder website title," [Online]. Available: <https://example.com>, 2026, accessed: Sep. 20, 2026.